ML-Enhanced Agent-Based Modeling Framework for simulating Food Insecurity Disparities in Philadelphia

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Abstract

New methodological approaches are needed to understand the discriminatory processes that drive persistent racial and ethnic disparities in house insecurity, food insecurity and dietary quality. This study introduces a unique framework combining an agent-based model (ABM) with machine learning and statistical tools to simulate the interactions of residential mobility, store choice, and diet in Philadelphia, USA. The model, shaped by qualitative input from stakeholders and tuned with empirical data, depicts how race/ethnic segregation, income inequality, and food environments interact to impact food access disparities. The results confirm that African American and Latino populations are overexposed to low-quality foods and food insecurity; this finding is consistent with observational evidence. The model allows for realistic simulation of potential policies, including policy scenarios—like modifications to eligibility and benefits levels of food and housing assistance programs—making for a useful tool. The framework can be applied to other urban contexts with sufficient data and provides lens through which to better understand food and housing related inequities and available policy levers in complex urban systems.

1 Introduction

Racial/ethnic disparities in diet and obesity are remarkably consistent across cities and have persisted or worsened over time [1]. In the United States, one-third of the population was obese in 2012 [2], and obesity prevalence was projected to increase across most states in the coming years[3]. Secular increases over the past decades have affected all age groups, genders, and racial/ethnic groups; however, the prevalence and increases in obesity have both been markedly unequal across these groups. In Philadelphia, obesity prevalence among Whites decreased from 29% in 2015 to 23% in 2018, increased from 32% to 39% among Hispanic or Latinos, and remained at 39-40% among African-American [4].

Obesity disparities largely reflect similar patterns in diet - one of the main behavioral factors influencing weight change [5, 6, 7, 8]. In Philadelphia, 41% of Hispanic or Latinos and 39% of African-American drink at least one sugar sweetened beverage per day, compared to 23% of Whites [9]. Additionally, 52% of Hispanic or Latinos, 42% of African-American, and 30% of Whites eat fast food at least once per week[9]. Nationally, data from the National Health and Nutrition Examination Survey (NHANES) show that African-Americans and Hispanic or Latinos have lower Healthy Eating Index scores than Whites, a measure of overall diet quality that is predictive of cardiovascular disease, cancer, and obesity [10, 11, 12, 13].

Food insecurity, defined as the lack of access to desirable, culturally appropriate, and nutritionally adequate food, is a key factor that influences healthy food intake and disproportionately affects racial/ethnic minorities [14]. Around 10% of households in the US were food insecure in 2019 [15]. Food insecurity is positively associated with unhealthy diets and, consequently, obesity, because financial constraints lead many households to turn to cheaper, less healthful and nutritionally deficient foods.

While the relationship between diet and food insecurity has been extensively studied [15, 16, 14], little evidence exists on the mechanisms that generate and sustain racial/ethnic disparities in food insecurity, nor on the dynamic relationships between food insecurity disparities and patterns in other health (e.g., diet, chronic disease) and social (e.g. residential segregation, income inequality) factors.

In particular, studies examining how these dynamics unequally affect racial and ethnic minorities remain limited. Existing conceptual frameworks have advanced the understanding of unhealthy diet and obesity risk as a complex, multi-scale problem influenced by individual, interpersonal, and environmental factors [17, 18, 19]. However, these frameworks do not specifically identify factors that disproportionately affect minorities and lead to disparities. There is a need for new, dynamic conceptual frameworks identifying the most important variables, relationships, and feedback loops that lead to worse outcomes among racial/ethnic minorities [20, 21]. A disparities framework can advance understanding of the social etiology of disparities and build evidence for policies that target the most important factors constraining food choice in minority communities.

Agent-based modeling (ABM) is a widely used tool for modeling dynamics involving individuals, their environment and the nonlinear interactions that emerge from these relationships [22, 23]. Computational models, including ABMs, have been used to support policy decisions related to diet and obesity. ABMs are particularly valuable for understanding how the same decision rules can lead to different outcomes among people with different characteristics or different contexts (e.g., in their home, work, and school environments). Despite the use of computational models and ABMs to study diet-related dynamics [24], significant opportunities exist to focus on disparities, specifically investigating population-level mechanisms, developing policy-relevant ABMs, and enhancing model validation through the use of widely available epidemiological and policy evaluation data.

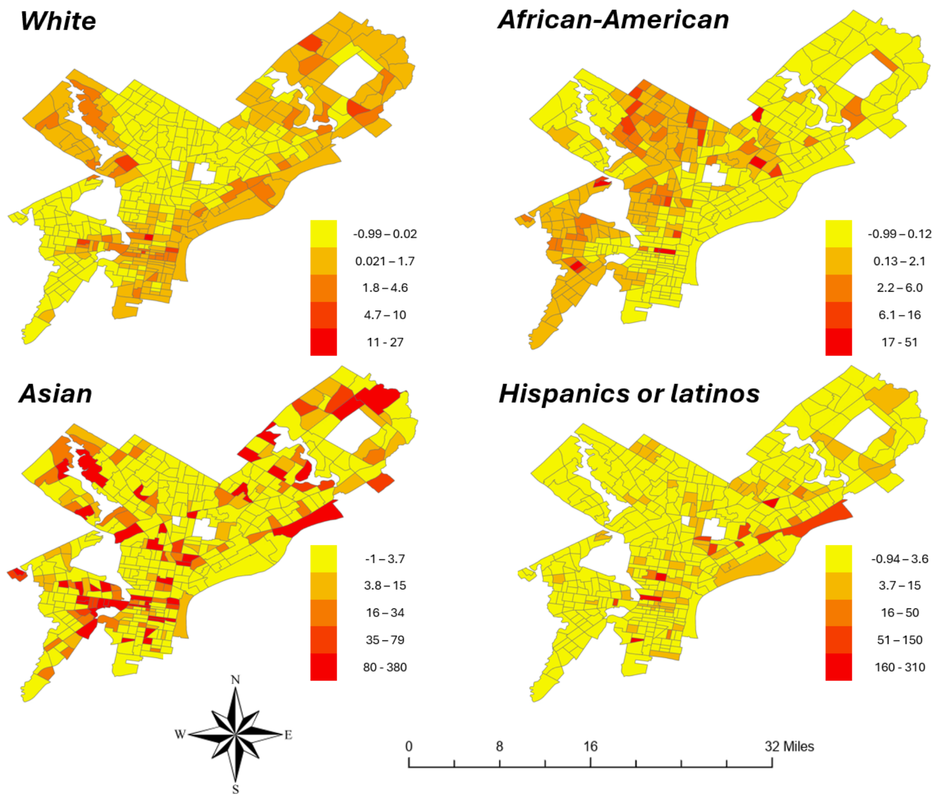
To capitalize on these opportunities, this study proposes a framework that integrates an ABM with various statistical and machine learning tools to analyze large-scale data and simulate dynamic interactions. This framework is informed by qualitative research with community, policy, and research stakeholders in Philadelphia, aimed at understanding the structure and function of complex systems that contribute to disparities in food security and healthy diets [25]. This research highlights the dynamic relationship between food affordability and the capacity to meet other non-discretionary costs, notably housing. The framework enables an examination of how racial/ethnic residential segregation [26], the inequitable distribution of food outlets [27, 28], the lower cost of energy-dense, non-nutritious foods [29, 30, 31], and income inequality [32] interact to shape racial/ethnic disparities in diet. Philadelphia serves as a case study to explore the interplay between food and housing insecurity.

2 Results

2.1 Characterization of synthetic population

A total of 646,604 households representing 1,433,522 persons were generated within the 408 census tracts in the Philadelphia area (N = 1,603,797 persons). The synthetic population's racial/ ethnic distribution closely matched census data, with average deviation of 2.1% at the census tract level. White populations were overestimated in some census tracts, with a maximum deviation of ~5.08%. Figure 1 presents a visual comparison between observed Census Bureau data and the synthetic population by race, highlighting the level of correspondence achieved.

Figure 1: Comparison of racial/ethnic distribution between census tracts between census and simulated data



From left to right and from top to bottom, the percentage difference is shown for Whites, African Americans, Asians and Hispanics or Latinos from census data vs simulated synthetic data.

To ensure computational efficiency and statistical representativeness, the sociodemographic characteristics of a sample of 100 households, distributed among the 408 census tracts (N = 38,682), was evaluated (Table 2). In cases where the total number of households in a census tract was less than 100, the total number of households available in that tract was used. This sample represents approximately 5.9% of the total households generated.

Table 1: Descriptive statistics that characterize the sample households

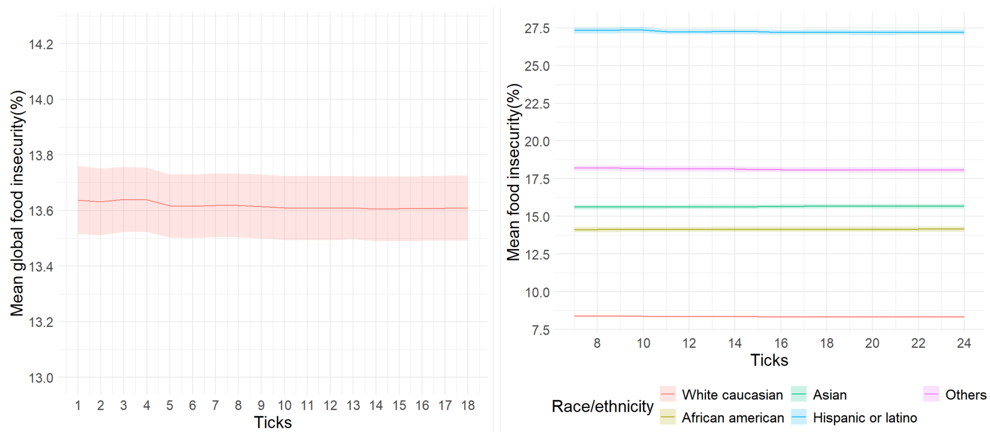
Household characteristics	**N = 38,682**[1]
Household type	
Female householder, no husband present	14,455 (37%)
Householder living alone	3,583 (9.3%)
Male householder, no wife present	9,835 (25%)
Married-couple family household	10,809 (28%)
Most prevalent race/ethnicity in the household	
African American	15,610 (40%)
White	15,217 (39%)
Hispanic or Latino	4,631 (12%)
Asian	2,260 (5.8%)
Others	964 (2.5%)
Household size (# members)	
1	14,455 (37%)
2-3	18,133 (47%)
4-5	5,032 (13%)
>6	1,062 (2.7%)
Household annual gross income (USD)	
<25K	10,698 (28%)
>=200K	2,789 (7.2%)
100K-150K	4,784 (12%)
150K-200K	1,886 (4.9%)
25K-50K	8,056 (21%)
50K-75K	6,141 (16%)
75K-100K	4,328 (11%)
Tenure of household	
Owner	21,638 (56%)
Renter	17,044 (44%)
Mean monthly rent (USD)	993 (sd: 866)

[1] n (%)

2.2 Calibration process

Food insecurity in the model is calibrated using food basket information and store choice, based on 20 simulation runs. This number was selected due to computational constraints, as increasing the number of simulations did not yield significant differences in the results. The calibrated model estimated an optimal store selection for 85.9% (CI: [85.2%,86.7%]) of agents. In addition, the simulated food insecurity rate varies between 13.5% and 13.75%, which aligns with the broader 13-20% range observed in Philadelphia. Figure 2 shows the simulated food insecurity at the city level and disaggregated by racial/ethnic group.

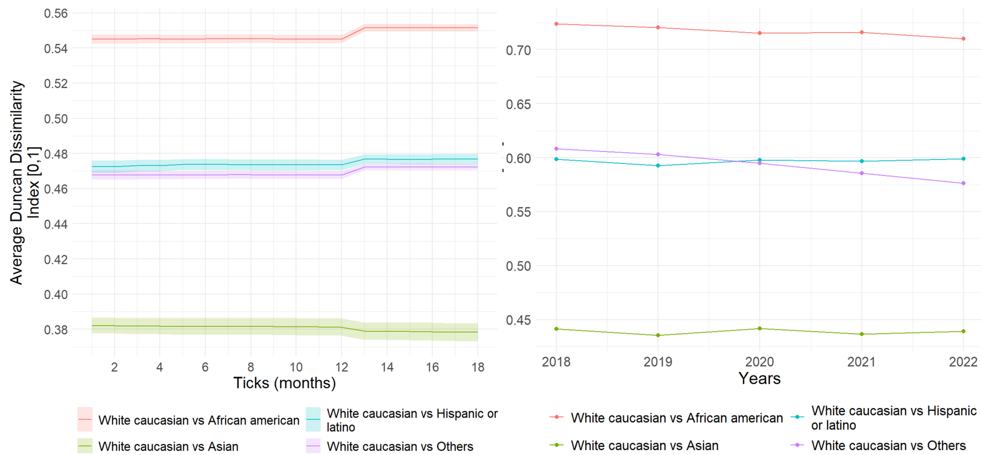
Figure 2: City-level food insecurity, and food insecurity distribution by race/ethnicity



The lines represent the average of the N runs and the band represents the 95% confidence interval.

The model replicates trends in residential segregation in Philadelphia based on ACS data (Figure 3). Although the lines do not exactly match the historical values calculated from census data between 2018 and 2022, they reflect the decreasing trend observed in that period, as well as the qualitative patterns observed between the different pairs of racial/ethnic groups: the highest segregation is between African-American and Whites, followed by Hispanics or Latinos and Whites, and the lowest between Asians and Whites.

Figure 3: Spatial segregation Duncan Dissimilarity Index distribution between racial/ethnic groups

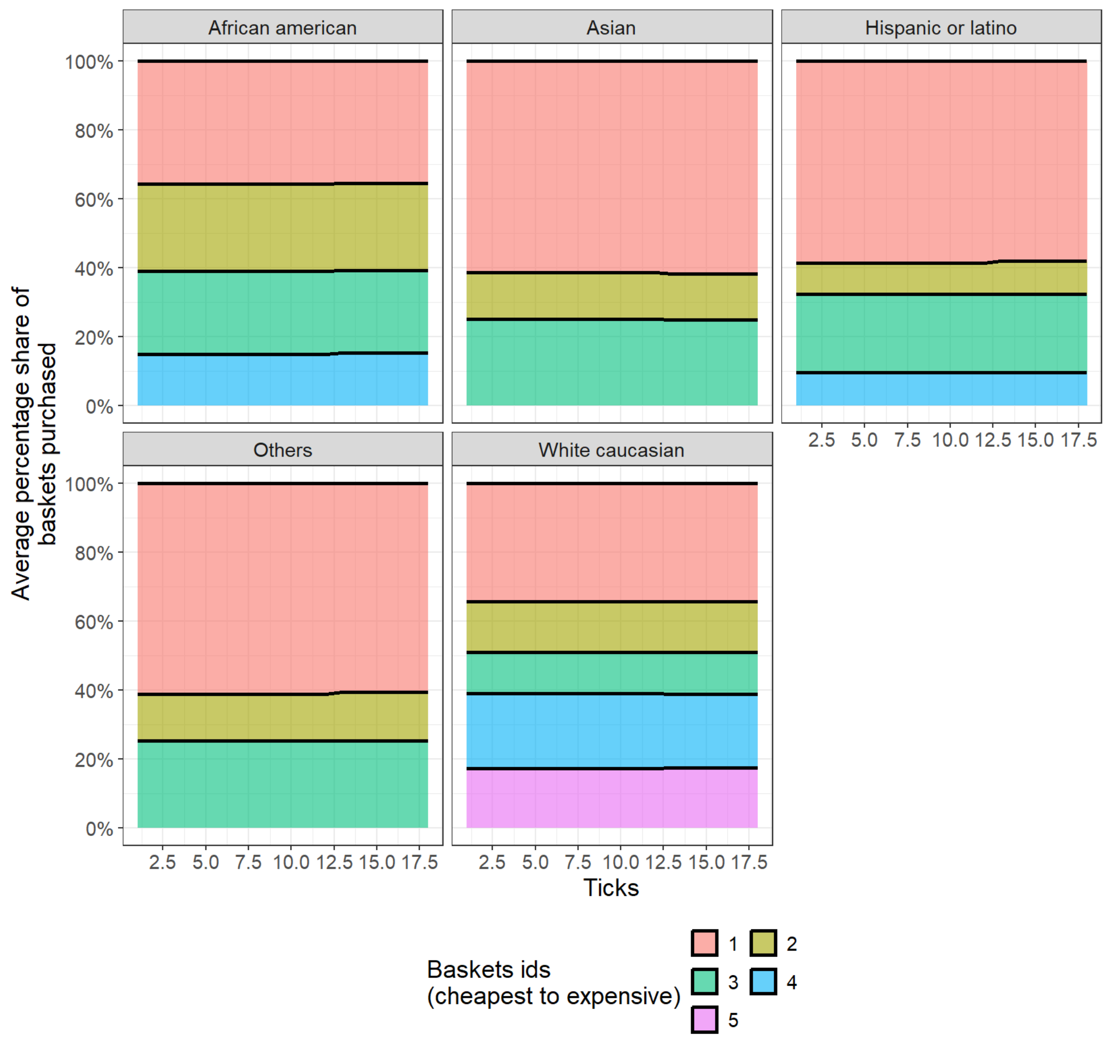


The lines represent the average of the N runs and the band represents the 95% confidence interval. On the left the data simulated with the model and on the right patterns observed in the ACS between 2018 and 2022.

2.3 Store selection and food basket

In the model, housing mobility can reconfigure the food basket assigned at initialization. The choice of stores is based on two main parameters: the number of establishments visited by each agent (3.59 on average according to the data versus 3.27 in the model) and the distribution of the type of stores visited, with supermarkets predominating. Figure 4 shows the simulated distribution of food baskets by race/ethnicity.

Figure 4: Distribution of food baskets by race/ethnicity



On average, Hispanics or Latinos, Asians and the "other" racial/ethnic group purchase lower quality baskets at the highest rates. The cheapest baskets tend to prioritize the consumption of ready-to-eat and ultra-processed foods. In contrast, Whites have the lowest reliance on the cheapest baskets, with a greater proportion of the population accessing more expensive, healthier baskets comprised of a higher proportion of protein, fruits and vegetables.

3 Discussion

We present a conceptual framework that integrates machine learning tools and statistical methods to capture, process, and generate data that inform a calibrated agent-based model to analyze disparities in food insecurity and dietary patterns in Philadelphia, a large city in the U.S. This city, characterized by marked racial/ethnic heterogeneity and a spatial distribution shaped by decades of segregationist policies, provides a key context for exploring how the combination of income inequality, food pricing, residential segregation and the unequal distribution of food retailers within cities work in combination to impact access to healthy food and, ultimately, diet quality. Our methodological approach complements previous advances in the study of unhealthy diets [17, 18, 19] by explicitly integrating environmental (housing stock, food retail) and social (e.g., population demographics) data to understand how social inequities by race/ethnicity [26] interact with urban environments to generate dynamics of food access and diet disparities [27, 28]. The model successfully reproduces observed patterns: residential segregation and income inequality are correlated with greater exposure to low-quality diets and food insecurity among African-American and Hispanic or Latino populations, consistent with findings from observational studies.

Although the rules incorporated into the model facilitated accurate calibration of food insecurity dynamics, additional calibration efforts focused on elements not directly governed by those rules—such as the maximum number of stores visited and the overall distribution of purchases across store types—helped ensure that the model produced outputs consistent with patterns reported in the literature [33, 34]. In parallel, the model also replicates patterns of spatial segregation that reflect the concentration of unhealthy food environments in neighborhoods with high racial/ethnic segregation and low income, an association widely documented in previous research [35, 36]. These structural patterns may be linked to historical processes of systemic discrimination [37, 38], such as redlining policies, which have shaped the unequal distribution of healthy food options in cities like Philadelphia.

This model further represents a methodological innovation by combining, for the first time to our knowledge, machine learning tools and advanced statistical methods to process data from multiple sources, generate synthetic populations, and represent complex dynamics linked to food insecurity. The strength of the proposed framework lies not only in the technical integration, but also in the active participation of community [39, 40], institutional and academic stakeholders, who contributed to define the rules governing the most relevant dynamics of the model [25]. This double dimension - technical and participatory - allows for a more accurate representation of complex urban realities. In particular, the model allows for a joint analysis of dynamics that have usually been addressed separately, such as residential mobility and the choice of food stores. In addition, it leverages detailed geographic information to more accurately estimate the costs associated with access to food baskets, the realistic location of residential units and the reproduction of spatial mobility patterns through robust algorithms [41].

This model represents an opportunity to simulate interventions and generate evidence to inform decisions aimed at reducing food inequalities, especially in urban contexts such as Philadelphia. The literature has documented the effectiveness of different policies-such as tiered taxes on unhealthy foods, regulation of the concentration of stores offering ultra-processed products, or incentives to locate healthy stores in low-income areas-in improving food security and reducing disparities in access to healthy foods [25, 42**?**]. Models such this allow us to represent different types of policies realistically, and to estimate their potential effects on key variables such as food insecurity or patterns of racial/ethnic segregation, at different geographic levels. In particular, this model incorporates rules that simulate participation in food assistance and public housing programs, which opens the possibility of evaluating the impact of these programs and communicating their relevance to local communities and decision makers.

This study has some limitations. First, like any ABM, our approach relies on specific assumptions about individual behavior and agent interaction, which, although informed by the literature and validated with local stakeholders, may not fully capture the complexity of real decisions. Second, data quality and availability limit the representation of certain key dimensions, such as individual food preferences or the exact nutritional quality of the products available in each store. However, the model manages to recreate reliable approximations of food baskets by race/ethnicity and income group, which represent the most disaggregated level possible with the available information. Furthermore, while the model is calibrated specifically for the city of Philadelphia, its results should not be automatically generalized to other urban contexts without careful adaptation to the social, economic and geographic particularities of each location. Finally, although advanced statistical tools and geo-referenced data were integrated, some technical decisions - such as the parameters used in the optimization algorithm for the choice of stores - may influence the results, so they should be interpreted as plausible approximations rather than exact representations of reality.

3.1 Conclusion

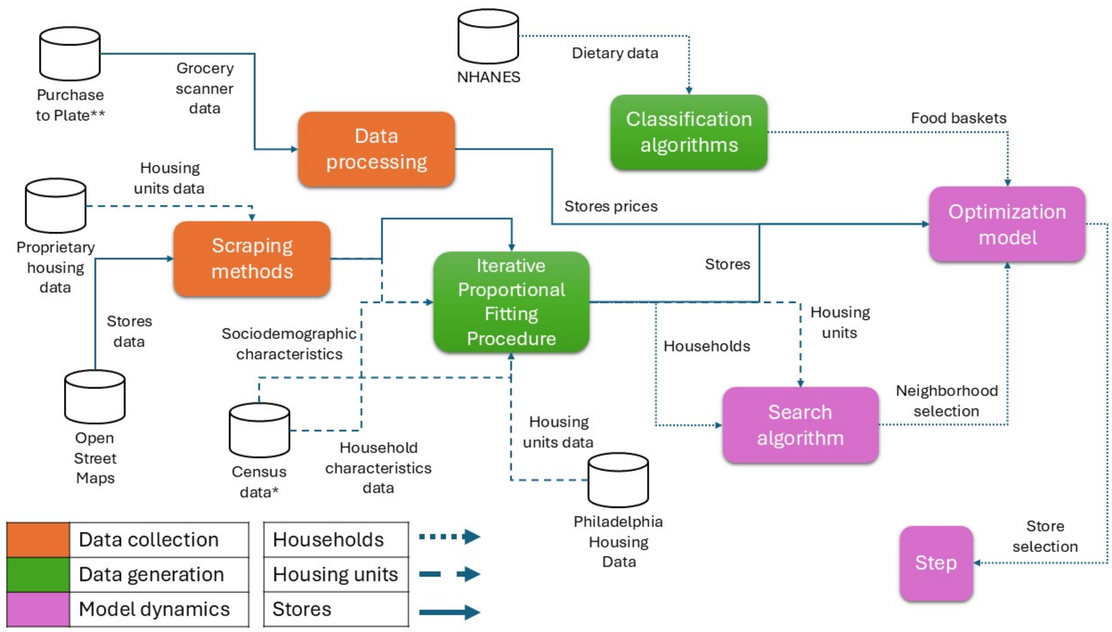
Residential mobility and dietary choice decisions are the result of complex processes involving multiple actors and their interactions with the environment. By designing a conceptual framework that combines advanced statistical and machine learning tools with an agent-based model, we demonstrate that it is possible to represent these dynamics at a high level of granularity, using data from a variety of sources. This methodological flexibility would allow us to adapt the model to other cities, provided that reliable data are available. In the case of Philadelphia, we note that race/ethnicity is a key factor in food insecurity dynamics. Our model highlights the importance of integrating residential mobility and store choice dynamics to more accurately represent food inequalities that disproportionately affect vulnerable groups. Policies such as the redistribution of healthy stores, the strengthening of food assistance and public housing programs, as well as incentives to increase participation in these programs, are channels through which decision makers can have an impact on reducing food insecurity and its unequal prevalence among racial/ethnic groups in the city.

4 Methods

4.1 Machine Learning framework

We propose a methodological approach that combines machine learning (ML) and optimization tools to process, transform, and integrate diverse data to inform the development of an ABM. This consists of three main phases, ensuring a structured and cohesive integration of techniques and information sources 5. The first phase focuses on data processing, where raw data is extracted, cleaned, and preprocessed to ensure quality and relevance. This stage uses extraction and preprocessing tools to prepare the data for subsequent transformations. In the second phase, both the processed data and additional datasets from external sources are transformed into structured inputs for the ABM using ML techniques. Here, ML is primarily used for imputation, estimation, and pattern recognition, rather than for simulating dynamics. The third phase involves integrating these structured inputs into the ABM to enable the operationalization of two key model dynamics: store and neighborhood selection. At this stage, ML and optimization tools are used to refine parameter calibration and support the ABM's simulation of interactions and processes, rather than directly generating synthetic data. Selection of variables included in the ABM was informed by input from community, policy, and research stakeholders involved in a group model building study to understand the structure and function of the systems that drive disparities in food insecurity and diet, as reported in [25].

Figure 5: Machine learning framework for the data and dynamics generation in the ABM



*Each color represents one of three phases:data collection (orange), data generation (green), and model dynamics (pink). Arrows represent different types of information flowing through the various process phases. Solid lines refers to flows of food stores data, round dotted lines represent flows of household data, dashed lines represent housing unit data. *Census data combines data from American Community Survey (ACS) and the Current Population Survey (CPS). ** Purchase to Plate Suite (PP-Suite) is a set of data products developed by linking household and retail grocery scanner data with the USDA Food and Nutrient Database for Dietary Studies (FNDDS).*

4.2 Data sources

Multiple sources were used to construct the different parts of the model presented in Figure 5. The American Community Survey (ACS) 2021 5-year sample, the Current Population Survey (CPS) 2021, and NHANES 2019-2020 were used as primary data sources to create a synthetic population of household-agents in "virtual Philadelphia". ACS was used to assign relevant household characteristics (e.g., household composition, race/ethnicity, income) with realistic geographic distribution across the city. CPS was used to assign participation in food assistance (e.g., the Supplemental Nutrition Assistance Program, or SNAP), housing assistance, as well as benefit values.

NHANES was used to characterize household's food baskets, which summarize household-level food purchasing across seven food groups: vegetables, proteins, grains, fruits, prepared foods, dairy products and other foods. We used dietary data from the 2017-2020 (n = 12,560), specifically, we used data from 24-hour dietary recalls to estimate mean daily consumption of the seven food groups among adults over 18 years of age with complete sociodemographic information (n = 7,896). Intakes in each food group were calculated as percentages of the total grams of food ingested and reported in the surveys. NHANES was also used to inform key variables in the store selection algorithm, which is used to determine the primary food stores at which each household' shops based on stores' inventories, quality, and prices.

Each household-agent is assumed to purchase food baskets composed of the following seven food groups: vegetables, proteins, grains, fruits, prepared foods, dairy products and other foods. We used dietary data from the 2017-2020 NHANES (n = 12,560). Specifically, we used data from 24-hour dietary recalls to estimate mean daily consumption of the seven food groups among adults over 18 years of age with complete sociodemographic information (n = 7,896). Intakes in each food group were calculated as percentages of the total grams of food ingested and reported in the surveys.

Data from the Philadelphia Housing Authority (PHA) and Redfin, a national real estate website, were used to create a synthetic population of housing units in Philadelphia. Specifically, the PHA data were refined using Redfin to ensure accurate pricing of the housing units used in constructing the synthetic population. The refined data was then used as the input for the Iterative Proportional Fitting Procedure (IPFP) to generate the synthetic sample of housing units.

To construct a population of food stores, we used data from OpenStreetMap (OSM) and official records on commercial establishments in the city from Neighborhood Food Retail dataset from the city of Philadelphia. To assign inventories and prices, we used data from the USDA's and the Purchase to Plate Suite (PPS). To generate the stores in the model, an exhaustive search was conducted using OSM through an extractor bot that scanned all zip codes in the Philadelphia area. This process collected information related to keywords such as "food", "supermarket", "store", "convenience store", and "grocery". Stakeholders involved in the larger food disparities project described pathways via which ethnic food stores (e.g., Asian Markets, carnicerias) are important food sources in racial/ethnic minorities communities that allow for cooking of culturally important heritage foods [25]. To identify ethnic food stores, the racial/ethnic group served, we used data from OSM (e.g., store names), official city sources (e.g., proprietor names), the demographic of zip codes, along with a custom classification algorithm to assign a race or ethnicity to each store. This procedure resulted in the creation of a database with 1648 stores distributed throughout the city.

For price estimation, the PPS database was used to calculate the means and standard deviations of the price per gram for each food group (proteins, dairy, fruits, vegetables, grains, prepared food, and others), stratified by race/ethnicity and income level to allow for sociodemographic differences in food purchasing within each broad food category. For each store, 500-meter buffers [43] were generated to identify the sociodemographics of households in each store's catchment area. The price/ cost of each households' purchases of a given food group draw from a normal distribution unique to each race/ethnicity, income group, and food group, where the mean and sd of this distribution were informed by the PPS data. The purpose of stratifying the distribution by race/ethnicity is a robust body of prior evidence suggesting that food purchasing patterns and diets vary across racial/ethnic groups [28, 44]. The money spent on purchases of a given food group by all households within a store's catchment i.e., 500-meter buffer, were used to inform the store's price distribution (including mean and sd) for each food group.

4.3 Iterative proportional fitting for household generation

In order to generate the synthetic population of households in Philadelphia, an IPFP was used that involved four steps [45]: variable selection, iterative proportional fitting, integerisation, and expansion. The first step consists of identifying the variables that adequately characterize households and enable the creation of a sample that closely reflects the distribution of the real population.

An essential part of this process is determining the level at which the data should be aggregated. Publicly available ACS microdata at the household level are only referenced geographically at the Public Use Microdata Areas (PUMA) level, which is quite a large spatial unit. Using the 1% sample available from the census, several variables were selected for analysis.

These included household type, which classifies households as either family or nonfamily. Family households include married-couple families, while nonfamily households are further divided into categories such as male or female householder without a partner, those living alone, and those not living alone. Household income was also selected, reporting the total income of all members aged 15 and older during the previous year, grouped into categories such as <25K, 25K–49.9K, and so on, up to 150K. Additionally, race was included, assigning the most prevalent race to each household. Categories for race include White, African-American, Asian (grouping Chinese, Japanese, and other Asian or Pacific Islander populations), and others (grouping American Indian or Alaskan Native, other races, and combinations of two or more races). Hispanic origin was not featured as a separate category because it is addressed in a different question in the ACS, and race statistics in the ACS do not classify Hispanic origin as a race. Aggregated statistics for household income, household type, and the intersection of household type and race were compiled from the 5-year ACS estimates at the census tract level to serve as inputs for estimating the weights needed to recreate the total population distribution.

To allocate individuals to census tracts, an iterative proportional fitting (IPF) algorithm was employed. This algorithm assigns a weight to each individual in a zone to reflect their representativeness. Weights are adjusted iteratively until the selected individuals replicate the distribution of the chosen variables for each zone. The result of this process is a set of fractional weights, representing the number of times each individual should appear in a given zone. However, since the selection process requires discrete integer values, representing whole individuals, these fractional weights must be converted into integers. This conversion process, known as integerisation, minimizes the loss of information by ensuring that high weights remain proportionally significant in the integerised version. As a result, individuals with higher weights are sampled more frequently, preserving the original distribution as accurately as possible.

The final step in the process, known as expansion, involves extending the database according to the number of times each individual should appear in each zone. Unlike typical IPF, this case is unique because individual data and aggregated statistics are available at different geographic levels—PUMA for individual data and census tract for aggregated statistics. To address this discrepancy, race-specific probabilities are calculated for each census tract, and households are assigned to the appropriate PUMA corresponding to each tract. This procedure continues until all households chosen to represent a PUMA are distributed among the census tracts. While some discrepancies between the actual and estimated distributions are inevitable, the average proportions of key variables, especially race—one of the most critical variables in this model—are preserved across the city.

4.4 Machine learning techniques for basket profile detection

We used k-means clustering with validation by silhouette analysis to identify common dietary patterns (basket profiles) within each racial/ethnic group. Identifying clusters within each racial/ethnic group, allows for cultural differences in diet between socially distinct groups. This stratification approach reflects the assumption that cultural and social norms around dietary preferences are more strongly shaped by race/ethnicity, while income primarily conditions access to specific consumption patterns. Therefore, clusters were defined within racial/ethnic groups, and income was incorporated later as a probabilistic modifier of cluster assignment.

Each observation from the dietary data was assigned to the most likely cluster corresponding to its race/ethnicity. In addition, averages and standard deviations were calculated for the servings in grams consumed per individual within each racial/ethnic group, which allowed us to estimate the overall monthly consumption per household, based on household size.

To account for socioeconomic differences, we used the ratio of household income to the Federal Poverty Line to classify observations into income groups. This allowed us to estimate the probability of different racial/ethnic and income groups belonging to specific consumption clusters or basket profiles. These probabilities are used to randomly assign households to consumption clusters, ensuring the model reflects the consumption patterns observed across income and by race/ethnicity groups.

4.5 The model

The model's key design elements are explained using the PARTE framework, which delineates agent Properties, Actions, Rules, Time, and Environment [46].

4.5.1 Agents properties

Table 2 describes the variables used for the agents (households):

Table 2: Data dictionary for household-level variables

Name	Description	Codification	Source
Census tract id	Census tract id as defined by U.S. Census Bureau.	Numeric	U.S. Census Bureau
Household serial number	Household serial number in the IPUMS.	Numeric	ACS2021 5yr
Household unique identifier	Unique identifier.	Numeric	Own (IPFP)
Race	Assigned based on the most prevalent race among household members.	0: White 1: African-American 2: Asian 3: Hispanic or Latino 4: Others	ACS2021 5yr
Household type	Household type constructed to match household census variable.	0: N/A 1: Married-couple family hh. 2: Male householder, no wife present. 3: Female householder, no wife present. 4: Male householder, living alone. 5: Male householder, not living alone. 6: Female householder, living alone. 7: Female householder, not living alone.	ACS2021 5yr
Total household income	Total income of all household members age 15+ during the previous year.	Numeric	ACS2021 5yr
Number of persons in hh	Number of persons in household calculated.	Numeric	ACS2021 5yr
Home ownership status	Indicates whether the household is a renter or owner.	0: N/A 1: Owned or being bought 2: Rented	ACS2021 5yr
Total monthly rent	Rent total price calculated from contract rent, electricity, gas, water, and heating fuel costs.	Numeric	ACS2021 5yr
Food basket profile	Food basket profile based on household income and race.	Numeric	ML basket profile (based on PPS database)

Definition of the acronyms
ACS2021 5yr: American Community Survey 2021 5-year estimates.
PPS: Purchase to Plate Suite database.
CPS: Current Population Survey.

4.5.2 Actions

Each month, agents make decisions related to their food purchases and whether to stay in their current housing unit. Both of these decisions are influenced by the decisions made in previous months. At the beginning of the month, agents that are housing insecure (households without sufficient income to pay their rent or mortgage) and that have not searched for a new home in the last three months begin searching for a new home using the neighborhood search process described in section 2.5.3.

Once the housing search is complete, an optimization model is used to inform the agent's search for a food store or set of stores that will allow them to purchase their household specific food basket for the month, given their food budget, the cost of the food basket at selected stores, and travel costs. If the agent does not find any affordable stores on the first attempt, they will increase their food budget allocation by up to 40% (if feasible) and again attempt to find an affordable store. If no affordable stores are identified despite this increase in budget allocation, the agent will search for a cheaper food basket cluster and repeat the process until they find an affordable basket.

If the cheapest food basket is unaffordable (i.e., its cost exceeds the agent's food budget allowance), the agent will seek to move to a cheaper housing unit (e.g., in a cheaper neighborhood, that conforms to their neighborhood preferences) where they can cover their food needs with a basic food basket to "free up" additional resources for food. The more expensive baskets have a higher proportion of proteins, vegetables and fruits, compared to the cheaper ones that have a higher proportion of ready-to-eat and ultra-processed products.

Finally, if none of these strategies proves feasible, the household will be classified as food insecure. At the end of the tick, a validation of the household's income and expenditures will be conducted to determine whether it should be classified as food insecure and/or household housing insecure. This will depend on its ability to afford rent, other associated costs (e.g., taxes, credits interests) and the allocation of at least a basic food basket.

4.5.3 Rules

Neighborhood selection

All agents perform an annual assessment of housing alternatives and may relocate to a neighborhood and housing unit that are better suited to their housing preferences. However, agents faced with food and housing insecurity can consider relocating to a new, more affordable neighborhood and housing unit each month, with the caveat that households cannot move more than once within a three-month period (this restriction was introduced to improve computational efficiency and corresponds to a period with no expected major changes.). As part of the neighborhood selection process, the agent evaluates the N feasible census tracts to identify census tracts with available housing. The utility (or desirability) of each of these tracts is determined by considering the neighborhood's racial/ethnic composition, median household income, and rental prices, relative to the agent's race/ethnicity and income [41]:

$$\begin{aligned}\text{Utility}_{ijt} = f(&\text{Neighborhood ethnic composition}_{jt}, \text{Ethnicity}_{it}, \text{Income}_{it},\\ &\text{Neighborhood 20th percentile rental price}_{jt},\\ &\text{Neighborhood median income}_{it}, \text{D}_{ijt}) \qquad (1)\end{aligned}$$

where D_{ijt} equals one if the potential neighborhood j in tick t for the agent i is the same as the original neighborhood for the agent. Estimates for the betacoefficients were used informed by [41]. The utility function was normalized to express each neighborhood's score from 0 to 1. The higher the score, the more likely it is that a given agent will relocate to the neighborhood. Since the utility array is ordered, if multiple neighborhoods have similar utility values, the first one in the array will be selected.

Housing unit selection

The non-developer agents explore housing units within their selected neighborhood, seeking properties that meet their requirements for the number of rooms and rental/sale price. Depending on whether the agent is a homeowner or a tenant, they will search for properties listed for sale or for rent, respectively. For simplicity, agents are either 'renters' or 'buyers', they do not switch from one to the other. If agents find more than one property that meets their requirements, one of these housing units is chosen at random. If the agent does not find any suitable properties that meet their criteria and fall within their budget, they will search for properties that meet their needs in the next most affinity neighborhood.

Real estate developers are an important part of the real estate landscape in Philadelphia and other cities. Among the reasons is that developers purchase for sale housing units and buildings, conduct renovations, and put the "flipped" units on the rental or sales markets. In the model, roughly 48.1% of housing units are owned by developers.

Developer agents (incorporated in the model as environment actors) calculate a sale price to rent revenue ratio, based on the expected market value of the property and its annual rent. If this ratio exceeds 15 (a reference value generally used in real state), the property is considered more profitable for sale than rent, prompting the developer to list the property for sale it after making modifications that increase its market value and monthly rent by 10%. If the rate is below 15, the developer chooses to rent the property at the last rented value.

Store selection

The model utilizes a multi-objective optimization approach to maximize affordability and utility based on the agent's budget, food needs, and store availability. The first step in the optimization model involves estimating the affinity between agents and stores. The affinity function calculates the average of the overall cost, ethnic affinity and variety of foods offered by the store, where the overall cost considers the average cost of a basket of food at each store, as well as the travel time estimated using the Euclidean distance between any given store and the agent's home. These costs are normalized ranging from zero to one, where stores with the lowest prices receive scores close to one. Ethnic affinity is a binary measure, where a score of one means that an agent's ethnicity matches the ethnicity of the store, while a score of zero signifies a mismatch. As for the store variety score, it takes a value of 1 if the store is a supermarket, 0.85 if it is a grocery store, and 0.7 for all other stores types (e.g. convenience stores, dollar stores, meat specialty).

Based on the affinity estimates and the average costs of the basket in each store, the model aims to maximize the affinity of the search with the established criteria. To achieve this, the following the optimization model is used:

$$OF : \sum_{i}^{I} \left(x_i \cdot aff_i - \frac{0.25 \cdot q_i \cdot aff_i}{food\_req} \right), \forall z \in Z \quad (2)$$

where x_i refers to a dichotomous variable that takes the value of 1 if store i is chosen and 0 otherwise for agent z, aff_i refers to the affinity estimated for store i for agent z, q_i refers to the purchased quantities in each store i for agent z. The objective function is designed with two main components that work together to optimize the model. The first component (first term in the formula) focuses on maximizing the affinity of the purchase made by the household, ensuring that the purchase decisions reflect the affinity criteria and thus the household's preferences. The second component (second term in the formula) seeks to balance the purchases made among the different stores, which is essential to avoiding very small purchases to stores with low affinity. This balancing not only ensures a more even and realistic distribution of purchases, but also reinforces the robustness of the model by considering both preferences and consistency in the distribution of purchases. The model seeks an optimal solution based on the below constraints:

$$R1 : \sum_{i}^{I} x_i \leq \text{max\_stores}, \forall z \in Z \tag{3}$$

$R1$ limits each agent's store visit visits to their individual maximum.

$$R2 : \sum_{i}^{I} q_i \cdot p_i \leq budget - \sum_{i}^{I} x_i \cdot oc_i, \forall z \in Z \tag{4}$$

$R2$ enforces a budget limit, which considers the he average cost for the basket p_i, and the commute cost oc_i for the agent z.

$$R3 : \sum_{i}^{I} q_i \geq food\_req, \forall z \in Z \tag{5}$$

$R3$ ensures that the household's monthly food requirements are met.

$$R4 : \quad q_i \geq x_i, \quad \forall i, z \in I, Z \tag{6}$$

$$R5 : \quad q_i \leq \frac{x_i \cdot food\_req}{max\_stores - 0.3}, \quad \forall i, z \in I, Z \tag{7}$$

$R4$ ensures that a purchase quantity q_i is only assigned if a store x_i is selected. $R5$ sets a maximum purchase quantity q_i unique to each store x_i. The model seeks an optimal solution based on these constraints. If the model cannot identify affordable stores that satisfy a given household's food purchasing needs, that household is classified as food insecure.

Time and environment

Each time step in the model corresponds to a month. Each month agents' make decisions choices concerning both their are updated and the previously described dynamics are simulated, including the choice of stores and housing and where they shop. The spatial environment of the model replicates the city of Philadelphia at real scale, using census tracts as the minimum geographic unit. Both stores and residential units are located in each of these tracts. Distance and travel time estimates are calculated using Euclidean distances. The spatial allocation of agents was performed according to the aforementioned geographic divisions, as described in Section 2.3.

4.5.4 Calibration process

To accurately represent the city's racial/ethnic composition, a calibration criterion was set. This criterion aimed to minimize the average absolute difference in the proportion of the four racial/ ethnic groups - Whites, African-Americans, Hispanics or latinos, and Asians - between the census data and the synthetic population generated for each census tract. The goal was to maintain these differences below 5%.

To calibrate the ABM, we focused on parameters related to store selection and internal mobility which could not be directly informed by empirical data. These include behavioral thresholds and weights used in the utility functions that govern agents' dietary choices and residential movements. These unknown parameters were varied over defined ranges in an iterative process to identify the set of values that best aligned the model's simulated outputs with observed trends in food insecurity and residential segregation at the city level.

Two variables were used as calibration targets. The first of these was food insecurity, which allows us to evaluate whether the model adequately captures the dynamics of store and dietary choice. Specifically, food insecurity rates estimated by Inguito [47] and Philabundance [48], ranging from 13% to 21.2% between 2006 and 2022, were used as empirical benchmarks. In addition to these aggregate values, the model sought to replicate observed disparities in food insecurity prevalence by racial/ethnic group, particularly the higher rates observed among African American and Hispanic populations [47]. The model was considered a good fit if its simulated food insecurity outcomes fell within a 2.5% of benchmarks.

The second calibration target focused on residential segregation, measured through Duncan's dissimilarity index [49], which was used to validate the spatial distribution patterns that emerge from agents' residential mobility. We calculated Duncan's dissimilarity index for Philadelphia, using data from the 2021 ACS. This index quantifies the level of segregation by comparing the distribution of two groups within specific geographic units relative to their overall distribution. Values range from 0 (complete integration) to 1 (total segregation) [49]. Based on our calculations, the highest dissimilarity values are observed between White and African-American populations (DD = 0.71), and between White and Hispanic / Latino populations (DD = 0.59). Due to variations in agent allocation during initialization, differences in the magnitude of the Duncan Dissimilarity index are expected. However, the model is considered adequately calibrated if the segregation hierarchy across the different group comparisons is maintained (see Figure 4).

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Data availability

Datasets are available from the corresponding authors upon reasonable request.

Authors contributions

All authors participated in all stages of the development of this article. All authors reviewed and approved the final manuscript.

Competing interests

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